CS4084
Quantum Computing
Week 04 – Lecture 07-08

Dr. Maqsood M. Khan

Department of Computer Science
FAST-NUCES Peshawar

Outline

1. Quick Review: What We Covered Last Lecture
2. Previously How Quantum States represented by Vectors & Standard Basis Measurement!
3. Exploring Classical States of Multiple Systems
4. Summary!

Quick Review: What We Covered Last Lecture

- **Quantum Information & Its Operations**
- **Dirac Notation (Third Part)**
- **Qubit Unitary operations**
- **Composing Unitary Operations**

Key Message: Dirac Notation provides a unified mathematical language that connects the physical state of a qubit to the Unitary operations that transform it and the Composition of complex quantum circuits.

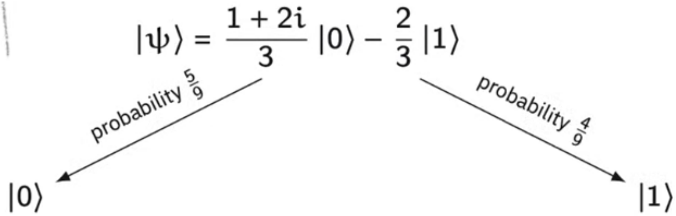
Quantum States represented by Vectors

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

$$|\psi\rangle = \frac{1+2i}{3}|0\rangle - \frac{2}{3}|1\rangle$$

Standard Basis Measurement



Unitary Operations

$$H\left(\frac{1+2i}{3}|0\rangle - \frac{2}{3}|1\rangle\right) = \frac{-1+2i}{3\sqrt{2}}|0\rangle + \frac{3+2i}{3\sqrt{2}}|1\rangle$$

Classical States

Suppose that we have two systems:

- X is a system having classical state set Σ.
- Y is a system having classical state set Γ.

Imagine that X and Y are placed side-by-side, with X on the left and Y on the right, and viewed together as if they form a single system.

We denote this new compound system by (X, Y) or XY.

Question

What are the classical states of (X, Y)?

Answer

The classical state set of (X, Y) is the *Cartesian product*

$$\Sigma \times \Gamma = \{(a, b) : a \in \Sigma \text{ and } b \in \Gamma\}$$

Continue..

The classical state set of the n-tuple $(X_1, \dots, X_n)$, viewed as a single compound system, is the Cartesian product

$$\Sigma_1 \times \cdots \times \Sigma_n = \{(a_1, \dots, a_n) : a_1 \in \Sigma_1, \dots, a_n \in \Sigma_n\}$$

Example

If $\Sigma_1 = \Sigma_2 = \Sigma_3 = \{0, 1\}$, then the classical state set of (X_1, X_2, X_3) is

$$\begin{aligned}\Sigma_1 \times \Sigma_2 \times \Sigma_3 = \{&(0,0,0), (0,0,1), (0,1,0), (0,1,1),\\ &(1,0,0), (1,0,1), (1,1,0), (1,1,1)\}\end{aligned}$$

Example

An n-tuple $(a_1, \ldots, a_n)$ may also be written as a *string* $a_1 \cdots a_n$.

Example

Suppose $X_1, \ldots, X_{10}$ are bits, so their classical state sets are all the same:

$$\Sigma_1 = \Sigma_2 = \cdots = \Sigma_{10} = \{0, 1\}$$

The classical state set of $(X_1, \ldots, X_{10})$ is the Cartesian product

$$\Sigma_1 \times \Sigma_2 \times \cdots \times \Sigma_{10} = \{0, 1\}^{10}$$

Continue..

Example

The classical state set of $(X_1, \ldots, X_{10})$ is the Cartesian product

$$\Sigma_1 \times \Sigma_2 \times \cdots \times \Sigma_{10} = \{0, 1\}^{10}$$

Written as strings, these classical states look like this:

$$\begin{array}{c} 0000000000 \\ 0000000001 \\ 0000000010 \\ 0000000011 \\ \vdots \\ 1111111111 \end{array}$$

Convention for Writing Cartesian Product!

Convention

Cartesian products of classical state sets are ordered *lexicographically* (i.e., dictionary ordering):

- We assume the individual classical state sets are already ordered.
- Significance decreases from left to right.

Example

The Cartesian product $\{1, 2, 3\} \times \{0, 1\}$ is ordered like this:

$$(1, 0),\ (1, 1),\ (2, 0),\ (2, 1),\ (3, 0),\ (3, 1)$$

When n-tuples are written as strings and ordered in this way, we observe familiar patterns, such as $\{0, 1\} \times \{0, 1\}$ being ordered as $00, 01, 10, 11$.

Summary: Key Takeaways

- **Unitary Evolution:** Quantum operations are reversible, norm-preserving transformations represented by 2×2 matrices for single qubits.
- **Composition of Operations:** Multiple gates applied in sequence correspond to matrix multiplication, where the algebraic order (right-to-left) must match the circuit order.
- **From Single to Multiple Systems:** Just as single systems have state sets, multiple systems are described by the combinations of their individual states.
- **Classical State Spaces:** For multiple classical systems, the combined state set is the **Cartesian Product** ($X \times Y$) of the individual state sets.
- **Notation & Convention:** We use string notation (e.g., xy or 01) as a shorthand for the ordered pairs in a Cartesian product to represent joint states.

Next Lecture: Moving from Classical Cartesian Products to Quantum **Tensor Products**, Entanglement, and the Postulates of Multiple